Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

medium: the material through which a mechanical wave travels

01 Bring back an earlier idea

A heat engine operates between 500 K and 300 K and absorbs 700 J.
Find maximum ideal work and explain why actual work is usually lower.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: v = fλ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at v=fλ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

Your turn: Draw both waves over the same 6 m length and label their frequencies.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two waves travel in the same ideal medium at the same speed. One has twice the frequency.
Compare wavelengths.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

04 Solve and justify

Sound travels at 340 m/s with frequency 425 Hz.
Find wavelength. Then find frequency for a .50 m wavelength in the same air.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student writes v=f/λ.
Check units and explain why that is not a speed.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

An object at 360 K sits in a vacuum facing surroundings at 240 K. How can it cool, and are absolute-zero surroundings required?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A wave travels 12 m in .40 s and has frequency 50 Hz.
Find speed and wavelength.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A wave enters a medium where speed changes from 12 to 8 m/s. Frequency stays 4 Hz.
Find both wavelengths and explain why frequency does not reset at the boundary.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3